$$P(u_2) = \left(\frac{\partial}{\partial x}[e^{-y}] + \frac{\partial}{\partial y}[e^{-y}]\right)^2 - [e^{-y}]^2 = 0 \quad \text{by (2) \& (1).}$$

$$\Rightarrow \left(0 - e^{-y}\right)^2 - e^{-2y} = e^{-2y} - e^{-2y} = 0 \checkmark, \quad \text{hence}$$

$u_2(x,y) = e^{-y}$ is a sol-n for (3), too.

Whereas, their linear combination (for superposition principle)

$u(x,y) = e^{x} + e^{-y}$ is NOT a sol-n for (3) because

$$P(u) = \left(\frac{\partial}{\partial x}[e^{x} + e^{-y}] + \frac{\partial}{\partial y}[e^{x} + e^{-y}]\right)^2 - [e^{x} + e^{-y}]^2 \overset{?}{=} 0$$

$$\Rightarrow (e^{x} - e^{-y})^2 - [e^{x} + e^{-y}]^2 = -4e^{x}e^{-y} \neq 0. \quad \text{So}$$

to answer if this violates Prin. of superp-n, recall Thm 5.5.9.

Let $P(\underline{x}, D) = \sum_{|\alpha| \le m} a^{\alpha} D^{\alpha}$ be a <u>linear operator</u>.

(for the <u>linear</u> PDE $\Rightarrow$ a^{α} coef-ts can NOT be a f-n of dependent variable)

Then, $P(c_1 u_1 + c_2 u_2) = c_1 P(u_1) + c_2 P(u_2)$.

Considering the given operator in (1) is NON linear, this operator can not be applied for Thm 5.5.9, hence the superpos-n principle is <u>not violated</u>. ∎

<u>Prob. 6.</u> (ZT, Ch.5, 9.2)

Let P be a <u>linear</u> PD operator.

(a) Show that if u_1 & u_2 are sol-ns to the nonhomogeneous eq-ns

(1) $$P(u_1) = f_1, \quad P(u_2) = f_2,$$

then $u = c_1 u_1 + c_2 u_2$ is a sol-n of the eq-n

$$P(u) = c_1 f_1 + c_2 f_2.$$

Sol-n: (a) [Preamble]

By Thm 5.5.9 given above, now we can apply this thm to the Operator P given, since it is linear. So

$$P(u) = P(c_1 u_1 + c_2 u_2) \overset{\text{by Thm 5.9.9}}{=\!=\!=} c_1 \cdot \underbrace{P(u_1)}_{f_1} + c_2 \cdot \underbrace{P(u_2)}_{f_2} \equiv c_1 f_1 + c_2 f_2 \checkmark$$

$\Rightarrow P(u) = c_1 f_1 + c_2 f_2$, showing $u = c_1 u_1 + c_2 u_2$ is a sol-n for the asked eq-n. ∎

(b). Suppose we have

(2) $P(xe^y) = x$ & $P(xy - y) = y^2$.

What's the sol-n to

(3) $P(u) = 3x + 4y^2$?

Sol-n: Again, P is linear & by Thm 5.9.9 then

$$P(u) = 3 \cdot \underbrace{x}_{\text{by (2), } P(xe^y)} + 4 \cdot \underbrace{y^2}_{P(xy-y)} = 3 \cdot P(xe^y) + 4 \cdot P(xy - y) =$$

P is linear

$$= P(3xe^y) + P(4(xy-y)) \overset{P \text{ is linear}}{=} P(3xe^y + 4(xy-y)), \text{ so}$$

$$P(u) = P(3xe^y + 4(xy-y)) = 3x + 4y^2, \text{ hence}$$

$u = 3xe^y + 4(xy - y)$ is a sol-n for (3). ∎

(c)

Let λ be a real # in an interval I of $\mathbb{R}^1$, & $\forall \lambda \in I$, the f-n $u(x, \lambda)$ satisfies the eq-n

(4) $$P(u(x,y)) = f(x,y).$$

Let $g(\lambda)$ be a f-n with properties listed below. Show that

(5) $$u(x) = \int_I g(\lambda)\, u(x,\lambda)\, d\lambda$$

is a sol-n to

(6) $$P(u) = f(x) \quad \text{where} \quad f(x) = \int_I g(\lambda)\, f(x,y)\, d\lambda$$

Assume that (i) we can interchange the partial derivatives that are used to define P & integration wrt (with respect to) λ over the interval I & (ii) $g(\lambda)$ is a f-n that gives us finite #s for both of the ↓ integrals:

$$\int_I g(\lambda)\, u(x,\lambda)\, d\lambda = \text{finite } \#$$

$$\int_I g(\lambda)\, f(x,y)\, d\lambda = \text{finite } \#.$$

sol-n: [Preamble]

Considering assumption (i), we can interchange $P(\underline{x}, D) = P\left(\begin{bmatrix} x \\ y \end{bmatrix}, D\right)$

$$P(u(x)) \overset{\text{by (5)}}{=} P\left(\int_I g(\lambda)\cdot u(x,\lambda)\, d\lambda\right) \overset{\text{by (i)}}{=} \int_I P\left[\underbrace{g(\lambda)}_{\text{const, not depends on } (x,y) = \underline{x}} \cdot u(x,\lambda)\right] d\lambda =$$

constant is pulled out as P is linear

$$= \int_I g(\lambda)\cdot \underbrace{P[u(x,\lambda)]}_{f(x,\lambda) \text{ by (4)}}\, d\lambda = \underbrace{\int_I g(\lambda)\cdot f(x,\lambda)\, d\lambda}_{\text{convergent, by assumption (ii)}} \overset{\text{by (6)}}{=} f(x).$$

Hence

$P(u) = f(x)$. ∎